

TED(15)-3023

Reg No.....

Revision 2015

Signature.....

THIRD SEMESTER DIPLOMA EXAMINATION IN MECHANICAL ENGINEERING

MANUFACTURING PROCESS

Time: 3 Hours

(Maximum marks :100)

I Answer all the questions in one or two sentences. Each question carry 2 marks

- 1 Define precision measuring instruments ?**
- 2 Describe least count of an instrument ?**
- 3 Define welding ?**
- 4 State unit cell ?**
- 5 List pattern materials? (5X2 = 10)**

PART—B

(Maximum marks: 30)

II Answer any five of the following question. Each question carries 6 marks.

- 1 How the measuring instruments are classified?**
- 2 Describe the working principle of a vernier caliper ?**
- 3. List the application of welding ?**
- 4. Sketch the important welding symbols ?**
- 5. Write down the advantages of forging ?**
- 6. What are the different types of pattern ?**
- 7. Describe the composition of molding sand ? (5X6 = 30)**

PART – C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

MODULE I

- III a) Sketch and explain an outside micrometer and mark its parts? 7**
b) What are gauges? List the important gauges? 8

OR

- IV a) Sketch and explain the working of dial indicator ? 7**
b) Discuss the advantages of pneumatic comparators ? 8

MODULE II

- V a) With neat sketch, explain the principle of arc welding ? 7**
b) Describe different oxy-acetylene gas welding flame with sketches ? 8

OR

- VI a) Explain TIG welding with proper figure ? 7**
b) What are welding defects? 8

MODULE III

- VII a) Explain various crystal defects ? 7**
b) Explain BCC, FCC & HCP ? 8

OR

- VIII a) List down the various forging methods and explain open die forging ? 7**
b) Differentiate hot drawing and hot piercing ? 8

MODULE IV

- IX a) Explain different pattern allowances? 7**
b) List and explain properties of molding sand ? 8

OR

- a) Differentiate between gravity die-casting and pressure die-casting ? 7**
b) Sketch and explain investment casting ? 8

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